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Final Task (Cart Total Calculator)

Code:

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<? cartTotalCalculator.html > ...
1  <!-- Problem Statement
2  You are building a shopping cart calculator. Each cart item is represented as an object with:
3  { id, name, price, qty, category }
4  The program must calculate:
5  1. Subtotal (sum of all items price × qty).
6  2. Discounts:
7  If quantity ≥ 3, apply 5% discount on that item.
8  If category = "stationery"; and subtotal of stationery items > 200, apply 10% category
9  discount.
10 3. Final Total after discounts.
11
12 Steps
13 1. Create the cart: An array of objects (3-5 products, different categories).
14 2. Use map() to calculate each item's total (price × qty).
15 3. Apply item-level discount using map() or inside reduce().
16 4. Use filter() to extract stationery items and calculate their subtotal.
17 5. Apply category-level discount if applicable.
18 6. Use reduce() to calculate overall subtotal and final total.
19 7. Use forEach() to print a formatted receipt.
20
21 Sample Input (Cart Example)
22 [
23 { id: 1, name: "Pen", price: 20, qty: 2, category: "stationery" },
24 { id: 2, name: "Mug", price: 150, qty: 1, category: "kitchen" },
25 { id: 3, name: "Notebook", price: 80, qty: 3, category: "stationery" }
26 ]
27
28 Expected Output (Approximate Format)
29 Item: Pen (x2) = 40
30 Item: Mug (x1) = 150
31 Item: Notebook (x3) = 240 + discount applied
32 -----
33 Subtotal: 430
34
35 Item Discounts: 12
36 Stationery Discount: 26.8
37 Final Total: 391.2
38
39 Reminders

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40 • Don't forget quantity × price when calculating totals.
41 • Discounts must be subtracted after subtotal is found.
42 • Use reduce() effectively to accumulate sums.
43 • Break the problem into small reusable functions.
44 • Format money properly (e.g., two decimal places).
45 Note: This is an implementation lab. Follow the hints carefully and write the full program
46 yourself. -->
47
48 <!DOCTYPE html>
49 <html>
50 <head>
51 <title>Cart Total Calculator</title>
52 </head>
53 <body>
54 <script>
55 // Part A - Setup the Cart
56 let cart = [
57   { id: 1, name: "Pen", price: 20, qty: 2, category: "stationery" },
58   { id: 2, name: "Mug", price: 150, qty: 1, category: "kitchen" },
59   { id: 3, name: "Notebook", price: 80, qty: 3, category: "stationery" }
60 ];
61
62 // Part B - Subtotal per Item
63 let itemTotals = cart.map(item => item.price * item.qty);
64 console.log("Item Totals:", itemTotals);
65
66 // Part C - Item-Level Discount (5% if qty ≥ 3)
67 let discountedItemTotals = cart.map(item => {
68   let subtotal = item.price * item.qty;
69   if(item.qty >= 3)
70     subtotal = subtotal - (subtotal * 0.05); // apply 5% discount
71   return subtotal;
72 });
73 console.log("Discounted Item Totals:", discountedItemTotals);

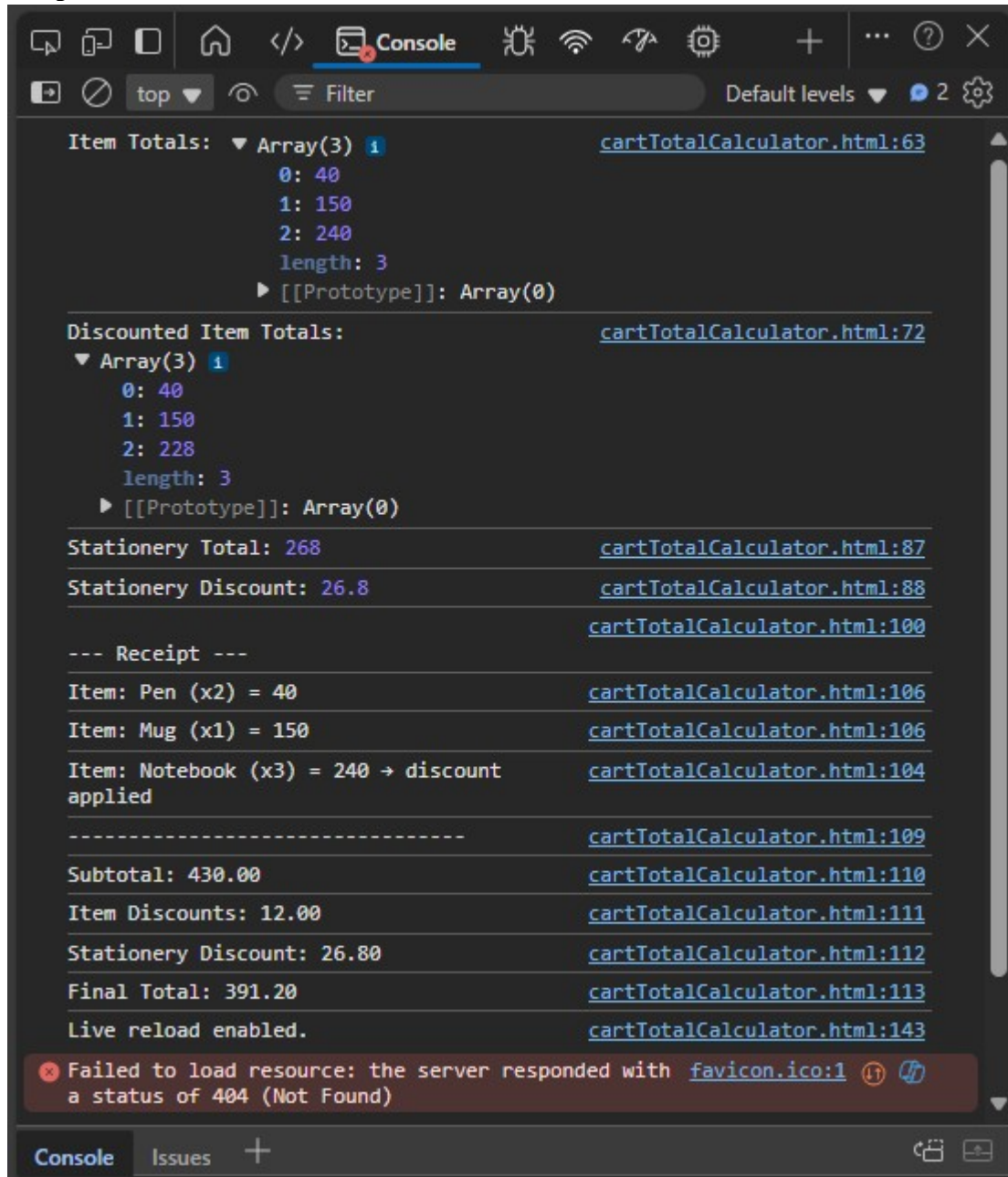
```

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74 // Part D - Category-Level Discount (10% on stationery subtotal > 200)
75 let stationeryItems = cart.filter(item => item.category === "stationery");
76 let stationeryTotal = stationeryItems.reduce((sum, item) => {
77   let subtotal = item.price * item.qty;
78   if(item.qty >= 3)
79     subtotal = subtotal - (subtotal * 0.05); //apply 5% discount first
80   return sum + subtotal;
81 }, 0);
82
83 let stationeryDiscount = 0;
84 if(stationeryTotal > 200)
85   stationeryDiscount = stationeryTotal * 0.10;
86
87 console.log("Stationery Total:", stationeryTotal);
88 console.log("Stationery Discount:", stationeryDiscount);
89
90 // Part E - Final Total
91 let subtotal = cart.reduce((sum, item) => sum + (item.price * item.qty), 0);
92
93 // Calculate total item-level discount
94 let totalDiscountFromItems = subtotal - discountedItemTotals.reduce((a, b) => a + b, 0);
95
96 // Final total = subtotal - item discounts - stationery discount
97 let finalTotal = subtotal - totalDiscountFromItems - stationeryDiscount;
98
99 // Print Receipt
100 console.log("\n--- Receipt ---");
101 cart.forEach(item => {
102   let lineTotal = item.price * item.qty;
103   if(item.qty >= 3)
104     console.log(`Item: ${item.name} (x${item.qty}) = ${lineTotal} + discount applied`);
105   else
106     console.log(`Item: ${item.name} (x${item.qty}) = ${lineTotal}`);
107 });
108
109 console.log("-----");
110 console.log("Subtotal:", subtotal.toFixed(2));
111 console.log("Item Discounts:", totalDiscountFromItems.toFixed(2));
112 console.log("Stationery Discount:", stationeryDiscount.toFixed(2));
113 console.log("Final Total:", finalTotal.toFixed(2));
114 </script>
115 </body>
116 </html>

```

Output:



```
Item Totals: ▼ Array(3) 1 cartTotalCalculator.html:63
  0: 40
  1: 150
  2: 240
  length: 3
  ▶ [[Prototype]]: Array(0)

Discounted Item Totals: cartTotalCalculator.html:72
▼ Array(3) 1
  0: 40
  1: 150
  2: 228
  length: 3
  ▶ [[Prototype]]: Array(0)

Stationery Total: 268 cartTotalCalculator.html:87
Stationery Discount: 26.8 cartTotalCalculator.html:88
cartTotalCalculator.html:100
--- Receipt ---
Item: Pen (x2) = 40 cartTotalCalculator.html:106
Item: Mug (x1) = 150 cartTotalCalculator.html:106
Item: Notebook (x3) = 240 → discount cartTotalCalculator.html:104
applied
----- cartTotalCalculator.html:109
Subtotal: 430.00 cartTotalCalculator.html:110
Item Discounts: 12.00 cartTotalCalculator.html:111
Stationery Discount: 26.80 cartTotalCalculator.html:112
Final Total: 391.20 cartTotalCalculator.html:113
Live reload enabled. cartTotalCalculator.html:143

❌ Failed to load resource: the server responded with favicon.ico:1
a status of 404 (Not Found)
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